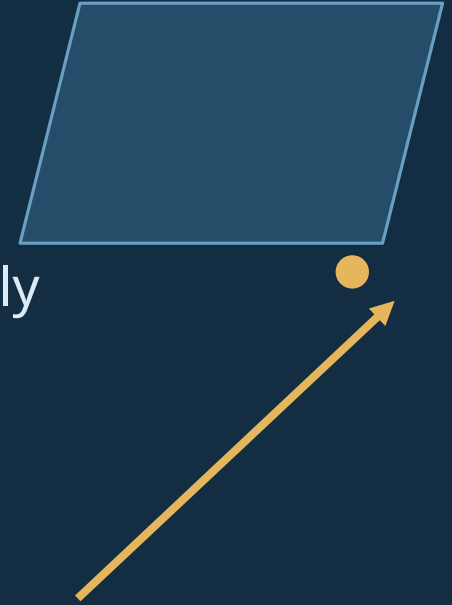


Estimability

Rank deficiency, estimable functions, and what the data can actually identify

$$Y = X\beta + \varepsilon$$



Learning objectives

By the end of this lecture, you should be able to...

ROADMAP

- Explain why rank deficiency makes the parameter vector nonunique.
- Define an estimable linear function $c^T\beta$ and recognize several equivalent criteria for estimability.
- Use the row space or null space of X to determine whether a parameter function is estimable.
- Distinguish nonunique coefficient estimates from unique fitted values.
- Explain the role of generalized inverses in rank-deficient linear models.

Why estimability?

The issue appears as soon as the design matrix is not full column rank.

MOTIVATION

$$E(Y) = X\beta$$

The data inform us about the mean vector $\mu = X\beta$.

If several different β vectors produce the same μ , then β itself cannot be uniquely identified.

Central question

Even when β is not identifiable, which linear combinations of β are uniquely determined by the model?

That is the problem of estimability.

Full rank versus rank deficiency

MOTIVATION Full column rank

One-to-one parameterization

$X\beta_1 = X\beta_2$ implies $\beta_1 = \beta_2$.

$$\mathcal{N}(X) = \{0\}$$

Every coordinate of β can, in principle, be identified.

Rank deficient

Nonunique parameterization

There exists $h \neq 0$ with $Xh = 0$, so $X(\beta+h)=X\beta$.

$$\mathcal{N}(X) \neq \{0\}$$

Only parameter functions that are constant along null-space directions can be identified.

A simple rank-deficient design

EXAMPLE

The third column is the sum of the first two:

$$x_3 = x_1 + x_2$$

Therefore $\text{rank}(X)=2$ even though X has 3 columns.

Interpretation

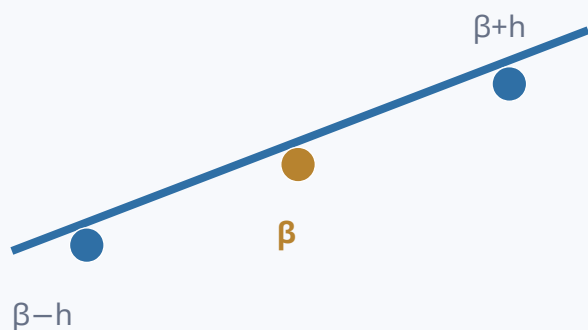
Moving β in the direction h changes the parameter vector but leaves the mean $X\beta$ unchanged.

Different coefficients, same mean

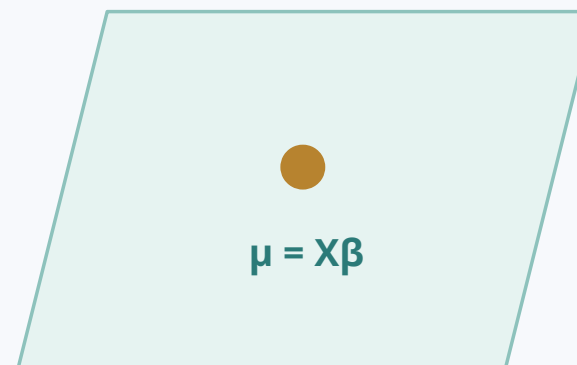
EXAMPLE

$$X(\beta + th) = X\beta + tXh = X\beta, \quad t \in \mathbb{R}$$

Parameter space



Mean space



Key idea

Rank deficiency is a problem of parameterization, not necessarily a problem of the mean model.

Definition: estimable linear functions

DEFINITION

A linear parametric function $c^T \beta$

is estimable if there exists a linear statistic $a^T Y$

such that $E(a^T Y) = c^T \beta$ for every β .

What this means

An estimable function is a feature of β whose value is uniquely encoded in the distribution of Y .

What it does not mean

The individual coordinates of β must be uniquely identifiable.

First criterion: solve $X^T a = c$

CRITERION

Start from the definition:

$$E(a^T Y) = a^T E(Y) = a^T X\beta = (X^T a)^T \beta$$

For this to equal $c^T \beta$ for every β , we need

$$X^T a = c.$$

Estimability criterion

$c^T \beta$ is estimable if and only if the linear system $X^T a = c$ has a solution.

The vector a supplies an unbiased linear estimator $a^T Y$.

Equivalent criterion: the row space of X

CRITERION

$$X^T a = c \iff c \in \mathcal{C}(X^T).$$

But $\mathcal{C}(X^T)$ is the row space of X.

Row-space criterion

A linear function $c^T\beta$ is estimable exactly when its coefficient vector c lies in the row space of X.

Practical interpretation

- Rows of X describe parameter combinations appearing in the mean structure.
- Estimable functions are exactly linear combinations of those rows.

Equivalent criterion: orthogonal to the null space

CRITERION

$$\mathcal{C}(X^T) = \mathcal{N}(X)^\perp$$

Therefore

$$c^T \beta \text{ estimable} \iff c^T h = 0 \text{ for every } h \in \mathcal{N}(X).$$

Most useful intuition

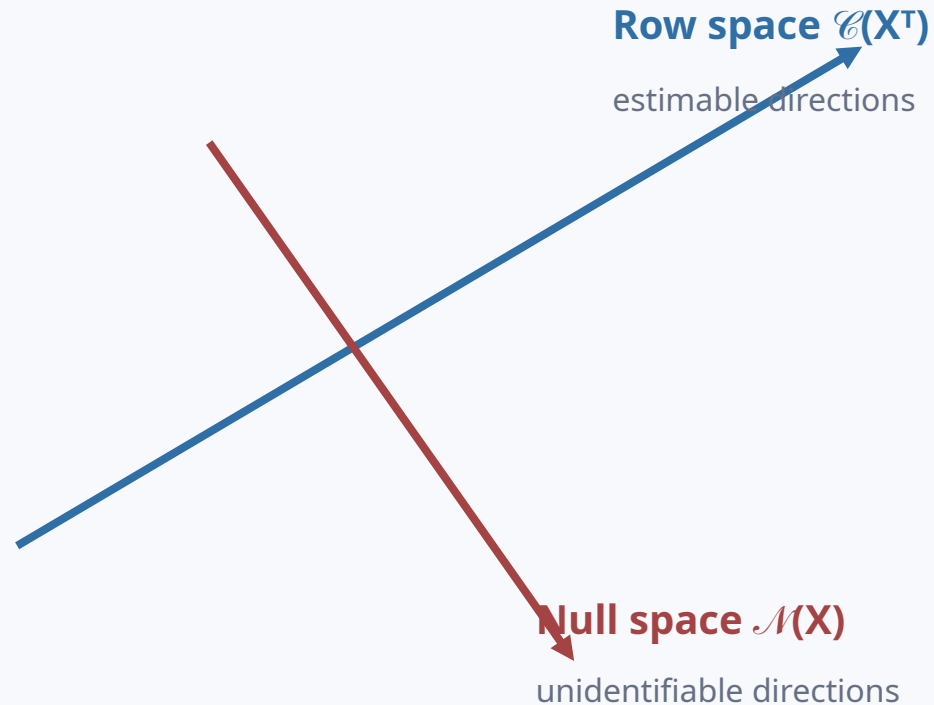
Directions h in the null space are invisible to the data. An estimable function must be unchanged when β moves in any invisible direction.

$$c^T(\beta + h) = c^T \beta + c^T h = c^T \beta$$

Geometry in parameter space

GEOMETRY

The parameter space decomposes into $\mathbb{R}^p = \mathcal{C}(X^T) \oplus \mathcal{N}(X)$.



What c must do

The coefficient vector c of an estimable function must lie entirely in the row-space component.

What c cannot do

It cannot have a component in a direction the model cannot distinguish.

A practical workflow for checking estimability

METHOD

- 1. Find the rank deficiency** Identify linear dependencies among columns of X .
- 2. Find null-space directions** Solve $Xh = 0$ for a basis of $\mathcal{N}(X)$.
- 3. Test the candidate function** For $c^T\beta$, compute c^Th for each null-space basis vector.
- 4. Interpret the result** If every product is zero, the function is estimable.

Worked example: find the invisible direction

EXAMPLE

From $x_3 = x_1 + x_2$,

$$x_1 + x_2 - x_3 = 0$$

so a basis for the null space is

Fast check

A candidate $c^T\beta$ is estimable exactly when $c^Th = 0$.

One dot product replaces repeated solving of $X^Ta=c$.

Worked example: which functions are estimable?

EXAMPLE

$$h = (1, 1, -1)^T$$

Function	c	c ^T h	Estimable?
β ₁	(1,0,0) ^T	1	No
β ₂	(0,1,0) ^T	1	No
β ₃	(0,0,1) ^T	−1	No
β ₁ + β ₃	(1,0,1) ^T	0	Yes
β ₂ + β ₃	(0,1,1) ^T	0	Yes
β ₁ − β ₂	(1,−1,0) ^T	0	Yes

The individual coefficients are not identified, but many contrasts and sums are.

The full estimable-function space

EXAMPLE

For the example, estimability requires $c^T h = c_1 + c_2 - c_3$

Therefore every estimable coefficient vector has the form

Dimension

The space of estimable functions has dimension $\text{rank}(X)=2$.

Basis

One convenient basis is $\beta_1 + \beta_3$ and $\beta_2 + \beta_3$.

How many independent estimable functions are there?

THEOREM

$$\{c : c^T \beta \text{ is estimable}\} = \mathcal{C}(X^T).$$

Therefore

$$\dim\{\text{estimable functions}\} = \text{rank}(X)$$

Interpretation

Rank(X) is not just the number of independent columns. It is also the number of independent pieces of linear information about β that the model can identify.

A familiar example: the one-way effects model

APPLICATION

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, 2.$$

Linear dependence

The intercept column equals the sum of the two group-indicator columns.

$$x_1 = x_2 + x_3$$

So rank(X)=2, not 3.

What is estimable in the one-way model?

APPLICATION

Parameter function	Meaning	Estimable?
μ	overall intercept parameter	No
α_1	group 1 effect parameter	No
α_2	group 2 effect parameter	No
$\mu + \alpha_1$	group 1 mean	Yes
$\mu + \alpha_2$	group 2 mean	Yes
$\alpha_1 - \alpha_2$	difference in group means	Yes

Lesson

Scientific quantities can be estimable even when the chosen parameterization contains nonidentifiable coefficients.

Estimability is invariant to arbitrary identifying constraints

INTERPRETATION

To report individual coefficients, software often imposes a constraint such as

$$\alpha_2 = 0 \quad \text{or} \quad \alpha_1 + \alpha_2 = 0.$$

Those constraints select one representative β from an entire equivalence class.

Important

A constraint can make the numerical coefficient vector unique, but it does not create information that was absent from the data.

Consequently

Any truly estimable function has the same value under every valid identifying parameterization.

When $X^T X$ is singular: generalized inverses

GENERALIZED INVERSE

If $\text{rank}(X) < p$, then $X^T X$ is singular and the usual formula

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad \text{is unavailable.}$$

A matrix G is a generalized inverse of A if

$$AGA = A.$$

For linear models

Take $A = X^T X$. Then G can be used to construct one least-squares solution.

A generalized inverse gives a least-squares solution

GENERALIZED INVERSE

$$\hat{\beta}_G = GX^TY$$

satisfies the normal
equations

$$X^TX\hat{\beta}_G = X^TY.$$

**Different generalized inverses can produce
different coefficient vectors.**

But...

The ambiguity lies only in null-space directions.
Those directions do not affect estimable functions
or fitted values.

Any two least-squares solutions differ only in a null-space direction

KEY RESULT

Suppose $\hat{\beta}_1$ and $\hat{\beta}_2$ both solve the normal equations.

$$X^T X (\hat{\beta}_1 - \hat{\beta}_2) = 0.$$

Because $\mathcal{N}(X^T X) = \mathcal{N}(X)$,

$$X (\hat{\beta}_1 - \hat{\beta}_2) = 0.$$

Therefore

$\hat{\beta}_1 - \hat{\beta}_2$ lies in $\mathcal{N}(X)$. The two coefficient vectors differ only in directions invisible to X .

Estimable functions are unique across least-squares solutions

KEY RESULT

If $\mathbf{c}^T \boldsymbol{\beta}$ is estimable, then $\mathbf{c} \perp \mathcal{N}(\mathbf{X})$.

For any two least-squares solutions, $\mathbf{h} = \hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_2$ belongs to $\mathcal{N}(\mathbf{X})$.
Hence

$$\mathbf{c}^T \hat{\boldsymbol{\beta}}_1 - \mathbf{c}^T \hat{\boldsymbol{\beta}}_2 = \mathbf{c}^T \mathbf{h} = 0.$$

Conclusion

Although $\hat{\boldsymbol{\beta}}$ itself may not be unique, the estimate of every estimable function is unique.

Fitted values are unique

KEY RESULT

Again let $\hat{\beta}_1$ and $\hat{\beta}_2$ be two least-squares solutions.

$$X(\hat{\beta}_1 - \hat{\beta}_2) = \mathbf{0}$$

immediately implies

$$X\hat{\beta}_1 = X\hat{\beta}_2.$$

The fitted mean is identified

There is one unique projection of Y onto $\mathcal{C}(X)$, even when there are infinitely many coefficient vectors representing that projection.

Nonunique coefficients $\neq$ nonunique fitted model.

Projection matrix in the rank-deficient case

PROJECTION

Let B be any generalized inverse of $X^T X$ and define

$$P = XBX^T.$$

Then P is the orthogonal projection onto $\mathcal{C}(X)$.

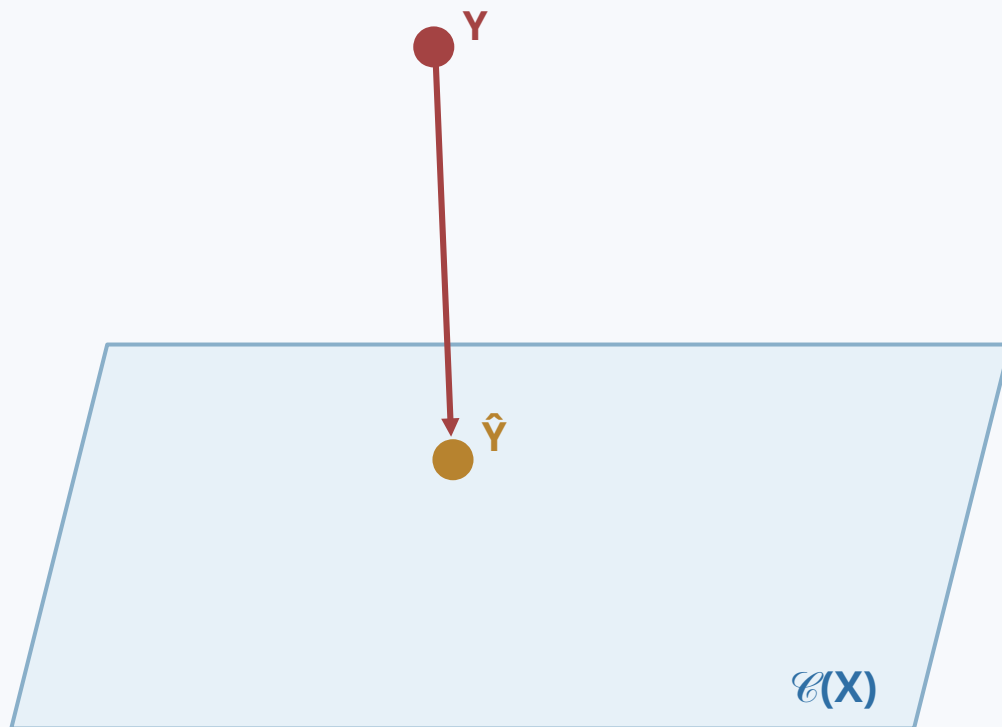
- $P^2 = P$ (idempotent).
- P acts as the identity on $\mathcal{C}(X)$.
- $Py = 0$ for $y \in \mathcal{C}(X)^\perp$.
- P does not depend on which generalized inverse B is chosen.

Same projection, many coordinates

Different generalized inverses may give different $\hat{\beta}$, but all produce the same $X\hat{\beta} = PY$.

The geometry is unchanged by rank deficiency

GEOMETRY



The least-squares problem still asks for the unique point in $\mathcal{C}(X)$ closest to Y .

What changes

The coordinates β used to represent that point need not be unique.

What does not change

The projected point $\hat{Y}$ itself is unique.

Common misconceptions

PITFALLS

"If β is not unique, nothing is estimable."

False. Rank(X) independent linear functions are estimable.

"A software constraint makes a parameter estimable."

False. It only chooses a parameterization.

"A generalized inverse uniquely estimates β ."

False. Different generalized inverses may produce different $\hat{\beta}$.

"Rank deficiency makes fitted values ambiguous."

False. The fitted mean $X\hat{\beta}$ is unique.

Practice 1

YOUR TURN

Without solving $X^T\mathbf{a}=\mathbf{c}$ separately for each item:

- Find a basis for $\mathcal{N}(X)$.
- Determine whether μ , α_1 , and α_2 are individually estimable.
- Determine whether $\mu+\alpha_1$, $\mu+\alpha_2$, and $\alpha_1-\alpha_2$ are estimable.
- Interpret each estimable function.

Strategy

Use the null-space dot-product test.

Practice 1: solution

SOLUTION

$$h = (1, -1, -1)^T$$

Function	$c^T h$	Estimable?
μ	1	No
α_1	-1	No
α_2	-1	No
$\mu + \alpha_1$	0	Yes
$\mu + \alpha_2$	0	Yes
$\alpha_1 - \alpha_2$	0	Yes

Interpretation

The identifiable quantities are the two group means and their contrast, not the arbitrary decomposition into μ and α parameters.

Practice 2

YOUR TURN

Suppose X has rank 3 but $p=5$.

- What is $\dim \mathcal{N}(X)$?
- How many linearly independent estimable functions of β are there?
- Could all five individual coefficients $\beta_1, \dots, \beta_5$ be estimable?
- Could $X\beta$ still be uniquely determined by β 's equivalence class?

Think structurally

You can answer all four questions without seeing the entries of X .

Practice 2: solution

SOLUTION

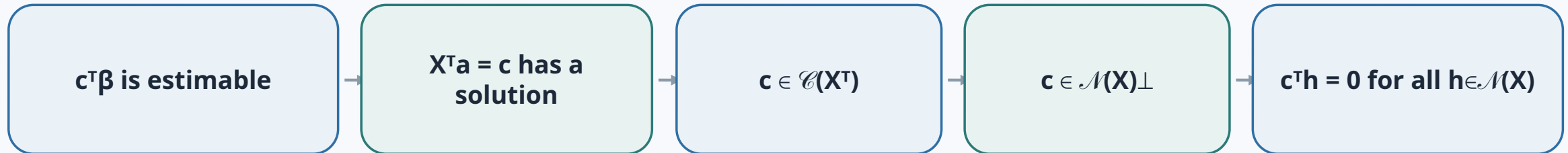
Question	Answer	Reason
$\dim \mathcal{N}(X)$	2	rank-nullity: $5 - 3$
Independent estimable functions	3	dimension of $\mathcal{C}(X^T)$
All 5 β_j estimable?	No	that would require 5 independent estimable directions
Is $X\beta$ identified?	Yes	all equivalent β values give the same mean

Big picture

Rank(X) measures the dimension of the mean model and the number of independent estimable parameter functions.

Estimability: the equivalence map

SUMMARY



Rank-deficient models

$\boldsymbol{\beta}$ may be nonunique, but every estimable function $\mathbf{c}^T \boldsymbol{\beta}$ and every fitted mean $\mathbf{X} \hat{\boldsymbol{\beta}}$ is uniquely determined.

Generalized inverses

They provide convenient coefficient representatives without changing estimable quantities or fitted values.

Takeaways

SUMMARY

- Rank deficiency means β is not uniquely identified, because X has nonzero null-space directions.
- Estimability asks which linear functions $c^T\beta$ are nevertheless uniquely determined by the mean model.
- The most useful tests are $c \in \mathcal{C}(X^T)$ and $c^Th=0$ for every $h \in \mathcal{N}(X)$.
- There are $\text{rank}(X)$ linearly independent estimable functions.
- Generalized inverses may produce different coefficient vectors, but estimable functions and fitted values remain unique.

Next: estimation and inference for estimable functions